

Creating Binary Circuits Through Tile Tumbling Burnout Bots

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Abstract

Full tilt systems are known for controlling large populations of robots, our Burnout Model has a unique limitation where it has only a certain amount of moves, afterwards it turns into an obstacle for other robots. Using these robots we use their limitations to create different forms of computation. By strategically using burnt-out bots to form pathways, we design and validate gadgets for universal logic gates, NAND and Fan-out, proving that the model is capable of universal computation.

1 Introduction

Full-tilt systems are systems that can employ a global force onto a given set of particles that can allow them to move in a unified direction and distance until reaching another particle or obstacle. Real world applications of such systems include robot populations where independent controls become less practical as the size of the robots decreases such as those at the nano-scale due to the limitations of the necessary hardware and the limited space to apply such components in respect to the robots' sizes[3]. We aimed to investigate how a game of *Burnout Bots*, modeled after the full-tilt system with the addition of a burnout condition, can be used to construct functional circuits with the implementation of universal logic gates: NAND and Fan-out. Burnout determines how many of these forces a particle can respond to before becoming inactive. Our work focused on overcoming such a constraint, given a grid of $m \times n$ tiles and a burnout amount of k to replicate the expected behavior of a given logical expression with deterministic commands that can be repeated in sequence.

2 Related Work

Previous work has investigated how global forces such as fluidic flow and electromagnetic fields can guide particles towards a target location[4]. Applications for such controls have been utilized in the context of drug delivery systems with blood vessels acting as a maze of obstacles[4]. Mobile particles exhibiting the full-tilt system behavior along with the introduction of *artificial walls*

as obstacles have been utilized for internal-logic problems where the particles themselves are part of a performed computation through the implementation of universal logic gates [3]. Using a dual-rail logic system, the limitation of a fan out gate was seen to have been evaded with a 2x1 framework of particles as opposed to a 1x1. Such circuits were composed and initially connected to variable gadgets which set the individual variables carried throughout. Channels that produced output that was ultimately not carried out were connected to a *waste receptacle* and remained unused for the remaining logic[3].

3 Methods

Due to the given unique constraint of burnout, we designated certain rules that mirrored those previously used in other related works.

3.1 Grid Environment

Let $G = \{0, 1, \dots, m-1\} \times \{0, 1, \dots, n-1\}$ be a finite 2D-grid where:

- W = Concrete Wall
- E = Traversable space
- B = Burnout Bot

3.2 Burnout Bot Model

Let k be the burnout value of a given bot. A burnout bot $b_i = (x_i, y_i, s_i)$ is a bot where:

- $(x_i, y_i) \in G$ is the bot's position at the current step
- $s_i \in \{k, k-1, \dots, 1, 0\}$ is the remaining available steps before burnout

Bot b_i is *burnt out* when $s_i = 0$ after which b_i cannot respond to global commands and is essentially considered a concrete wall and thus:

- $b_i \in W$

Let the set of valid global commands for a given $b_i \in B$

- $d = \{U \uparrow, D \downarrow, L \leftarrow, R \rightarrow, Spawn\}$

where '*Spawn*' creates bots at all spawn locations in G , and UDLR are movement commands in which the bot b_i moves maximally in that direction.

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3.3 NAND Circuit Encoding

Theorem 1 *Direction Sequence Universality for NAND Gate with Burnout*

To encode the initial input variables of the circuit, we determined that a variable gadget would be necessary in order to retain the order of our directional sequence. This gadget works in combination with our NAND gadget. The global sequence of commands for all bots in our constructed NAND implementation is given by:

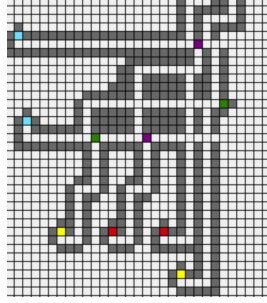


Figure 1: NAND Gadget

- $\{D \downarrow, R \rightarrow, U \uparrow, R \rightarrow, U \uparrow, \text{Spawn}\}$

the repetition of which will force the bots to perform calculations encoded within their positions in G. We used the ASARG TumbleTiles Software to evaluate the provided sequence, assuming a burnout value of $k=5$. Our variable gadget is composed of 4 incoming channels that feed into a 2-input NAND utilizing dual-rail logic to represent the 4 potential bit combinations of a given set of arbitrary variable combinations A and B. As seen in Fig. 1, a burnout of 5 results in 4 various idle positions throughout the grid, color coded as purple and green Burnout Bots. After which, the bots can no longer move in accordance to a received global command. Once this occurs, we can direct the navigation of a newly spawned bot which uses the burnt out bots as a barrier to accurately represent the present output from the previous inputs. These outputs are then used for the following present gadget.

Theorem 2 *Universal Direction Protocols for Fan-Out Circuits*

Furthermore, the implementation of universal circuits requires the ability to have outputs from a NAND circuit span into several other inputs to be fed into additional NAND circuits. We designed a fan-out gadget that would behave in such a manner through the previously defined sequence mentioned in Theorem 1. From this we expect to be able to construct any given logical expression, by combining the results from our two gadgets. The fan-out gate allows us to branch any arbitrary variables A and B into T number of inputs where $T \in \mathbb{Z} \geq 2$.

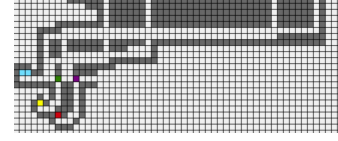


Figure 2: Fan Out Gadget

Utilizing the same repetitive sequence stated earlier, the fan out gadget takes one set input and works to branch it into T-fanouts. The fan out gadget utilizes a buffer to set bots into designated positions (similar to how they acted in the NAND gadget) to where they burnout and allow the newly spawned bots to properly navigate to their coinciding input tunnels. The tunnels follow the format of 1001 from left to right, where these would match the coinciding inputs into either another Fanout Gadget or a NAND gadget.

4 Our results

Through the testing of both our gadgets, we were able to test the theoretical validity and limitations that they held. We found that if given a predetermined sequence of gadget combinations and/or a logical equation, it is indeed possible to produce the correct output. We were able to produce a connection sequence of a NAND gate device followed by a fan-out device with $T = 2$. We were successfully able to achieve the desired outputs of 0 with inputs of 0/0 into the NAND system.

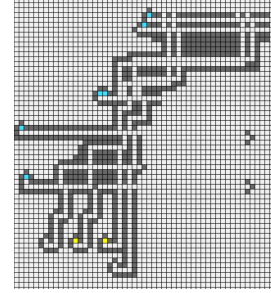


Figure 3: NAND leading into Fanout Gadget Circuit

4.1 Limitations

Throughout the process, we found that the model held some limitations within its design and execution. We had to ensure that all circuit logic could properly be implemented with the burnout constraint. Additionally, when creating combination circuits, the size of the circuits could become so large that the complexity would not merit the circuits' benefits. Furthermore, we found that coinciding specific outputs with continuing gadget inputs could easily get mixed and produce wrong outputs.

5 Conclusion

We have successfully demonstrated that our limited Full Tilt Burnout Bot is capable of performing logical computations. This allows us to improve the capabilities and limitations of these systems and also allows us to reduce our findings to real-world scenarios like nano-robotics, programmable matter, or swarm systems where components might fail. Future work should focus on optimizing the spatial efficiency of the gadgets. A theoretical analysis of the model's computational complexity is also a promising direction.

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